Fertigation Unit



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Nutrient pump/ Fertigation Unit

What it is used for

A fertigation unit delivers dissolved fertilisers through an irrigation system to supply nutrients to crops in measured quantities. Correct nutrient preparation, injection-rate control, and irrigation monitoring help improve nutrient distribution while reducing clogging, crop injury, and environmental contamination.

Main parts

- Fertiliser or nutrient mixing tank
- Injection pump or dosing device
- Irrigation water inlet and filter
- Flow meter, pressure gauge, and control valves
- Injection line and non-return valve
- Controller, hoses, and distribution network

Before you start

1. Confirm the crop, growth stage, irrigation schedule, water quality, and approved nutrient recipe.
2. Read the unit manual and fertiliser labels to verify compatibility, mixing order, and required concentration.
3. Inspect the tank, pump, filters, hoses, valves, injection line, and electrical connections for leaks or damage.
4. Check that the irrigation filter is clean and the non-return valve or backflow protection is correctly installed and functional.
5. Prepare fertiliser solution using the specified quantities and safe handling procedures; do not mix incompatible chemicals.
6. Verify that the pump, flow meter, pressure gauge, and controller are functioning and that the field irrigation system is ready.

How to use

1. Place the unit on a stable surface and confirm that the mixing tank is clean and suitable for the nutrient solution.
2. Measure and add water to the tank according to the approved mixing procedure.
3. Add the specified fertilisers in the recommended order while mixing; ensure each component dissolves as instructed.
4. Close the tank securely and inspect the prepared solution for undissolved material or unexpected precipitation.

5. Open the required irrigation valves and start the water flow according to the system's startup procedure.
6. Confirm that irrigation pressure and flow are within the permitted operating range before nutrient injection.
7. Set the dosing rate, target concentration, and injection duration according to the crop plan and equipment instructions.
8. Start the injection pump and monitor solution level, water flow, pressure, and dosing indicators.
9. Check the distribution system for leaks, blocked emitters, unusual pressure changes, or uneven irrigation.
10. Adjust settings only within approved limits if the measured dosing rate differs from the required value.
11. Stop nutrient injection if a leak, backflow risk, pump fault, or unexpected pressure change occurs.
12. At the end of dosing, stop the injection pump and flush the irrigation lines with clean water according to the approved procedure.
13. Close valves in the prescribed sequence and prevent residual concentrated fertiliser from entering water sources or drains.
14. Record nutrient type, concentration, applied volume, field or crop area, and any operating issues.

⚠ Safety

- Only trained personnel should operate and maintain the fertigation unit.
- Wear chemical-compatible gloves, eye protection, and protective clothing when preparing or handling fertiliser solutions.
- Follow product labels and safety data sheets; never mix incompatible fertilisers or add chemicals without an approved procedure.
- Protect drinking-water supplies against backflow using suitable approved prevention devices.
- Never open pressurised lines or remove filters while the system is operating; isolate pressure first.
- Keep electrical components dry and isolate power before servicing pumps or controls.
- Prevent spills and dispose of leftover solution and rinsing water according to product instructions and local environmental requirements.

Care & storage

- Rinse the mixing tank, injection lines, and pump according to the fertiliser and equipment instructions after use.
- Clean irrigation filters and inspect emitters, valves, and non-return devices regularly.
- Check pump seals, hoses, fittings, and electrical connections for wear or leakage.
- Verify dosing accuracy and flow-meter readings at the prescribed intervals.
- Store fertilisers safely and arrange servicing if injection rate, pressure, or pump operation becomes unstable.

Common mistakes to avoid

- Using the wrong nutrient concentration or mixing incompatible fertilisers.

- Starting injection before confirming irrigation flow and pressure.
- Ignoring clogged filters, leaks, or backflow protection failures.
- Failing to flush the system after dosing or disposing of concentrated solution improperly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.